GeoRAG System

Query Evaluation Documentation

Location-Aware Retrieval-Augmented Generation for Karlsruhe

System	GeoRAG
Location	Karlsruhe, Germany
Application Runtime	Local Python CLI
Vector Database Runtime	Docker Compose (Milvus standalone + etcd + minio)
Embedding Model	all-MiniLM-L6-v2
Vector DB	Milvus

1. System Overview

GeoRAG is a location-aware Retrieval-Augmented Generation (RAG) system for queries about places in Karlsruhe. It combines semantic retrieval from a Milvus vector index with location-aware reranking and generates route visualizations in HTML format.

1.1 Architecture

The system processes queries in these stages:

1. User enters a natural-language query and optional starting location (address or lat/lon).
2. Location input is resolved to coordinates (or falls back to default Marktplatz if unresolved).
3. Query text is embedded using all-MiniLM-L6-v2.
4. Milvus retrieves top semantic candidates.
5. Candidates are reranked by hybrid score (semantic relevance + distance).
6. Response is generated:
    - LLM response if Ollama/Mistral is available and has enough memory
    - Fallback formatted response if LLM generation fails
7. Route map is generated and saved as interactive HTML.

1.2 Deployment

The Python application runs locally from the command line. Milvus infrastructure runs via Docker Compose containers. This separation keeps vector search stable while allowing local model testing.

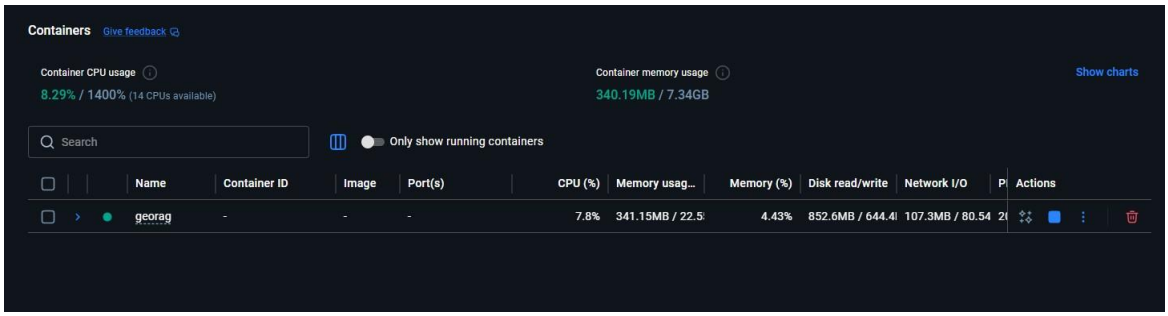

Figure 1: Docker container running the GeoRAG system

2. Query 1 – Shopping Type Query

2.1 Query Details

Original Query	I want to buy a new washing machine and where can I go to check in the city?

User Location	input-based; if unresolved, fallback to Marktplatz, Karlsruhe.

2.2 System Execution

The user ran the query from a terminal with the following command:

```
(venv) PS C:\Users\boopo\OneDrive\Desktop\New folder\georag> python main.py ask " I want to buy a new washing machine and where can I go to check in the city?"
Where are you at right now? Enter address or lat,lon (press Enter for default Marktplatz): mannheim karlsruhe
============================================================
  GeoRAG - Query
============================================================
  Question:  I want to buy a new washing machine and where can I go to check in the city?
  Location: PSD Bank Karlsruhe-Neustadt eG, Bismarckstraße, L15, Innenstadt, Innenstadt/Jungbusch, Mannheim, Baden-Württemberg, 68161, Deutschland
============================================================
Loading embedding model (all-MiniLM-L6-v2)…
  (Downloads ~90 MB on first run – cached afterwards.)
```

Figure 2: Terminal execution – Shopping query input

2.3 Retrieved Context

Top retrieved candidates were mostly supermarkets/department stores (for example REWE City, Woolworth). This indicates partial relevance but also shows missing appliance-specialized entries in the indexed POI set.

```
--- Original Query ---
 I want to buy a new washing machine and where can I go to check in the city?

--- Retrieved Context ---
[1] REWE City  (Shopping)
    Address  : Ettlinger-Tor-Platz 1, Karlsruhe
    Hours    : Mo-Sa 08:00-20:00; Su,PH off
    Website  : https://www.rewe.de/marktseite/karlsruhe/840255/rewe-markt-ettlinger-tor-platz-1/
    Score    : 0.3450
    Snippet  : REWE City is a shopping venue, located at Ettlinger-Tor-Platz 1, Karlsruhe. It is a supermarket selling groceries and household items. Opening hours: Mo-Sa 08:0...

[2] REWE City  (Shopping)
    Address  : Kaiserstraße 227, Karlsruhe
    Hours    : Mo-Sa 07:00-24:00
    Score    : 0.3364
    Snippet  : REWE City is a shopping venue, located at Kaiserstraße 227, Karlsruhe. It is a supermarket selling groceries and household items. Opening hours: Mo-Sa 07:00-24:...

[3] Woolworth  (Shopping)
    Address  : Kaiserstraße 148, Karlsruhe
    Hours    : Mo-Sa 09:00-20:00; Su,PH off
    Score    : 0.3338
    Snippet  : Woolworth is a shopping venue, located at Kaiserstraße 148, Karlsruhe. It is a department store offering a wide range of goods. Opening hours: Mo-Sa 09:00-20:00...

[4] Netto City  (Shopping)
    Address  : Otto-Wels-Straße 33, Karlsruhe
    Hours    : Mo-Sa 07:00-21:00; PH closed
    Website  : https://www.netto-online.de/filialen/Karlsruhe-Oberreut/Otto-Wels-Str-33/3032/
    Score    : 0.3261
    Snippet  : Netto City is a shopping venue, located at Otto-Wels-Straße 33, Karlsruhe. It is a supermarket selling groceries and household items. Opening hours: Mo-Sa 07:00...

[5] Woolworth  (Shopping)
    Address  : Rheinstraße 22, Karlsruhe
    Hours    : Mo-Fr 09:00-19:00; Sa 09:00-18:00
    Score    : 0.3228
    Snippet  : Woolworth is a shopping venue, located at Rheinstraße 22, Karlsruhe. It is a department store offering a wide range of goods. Opening hours: Mo-Fr 09:00-19:00; ...
```

Figure 3: Top-5 retrieved context documents for the shopping query

Summary of retrieved candidates:

- [1] REWE City – Ettlinger-Tor-Platz 1, Karlsruhe (Score: 0.3450) – Supermarket
- [2] REWE City – Kaiserstraße 227, Karlsruhe (Score: 0.3364) – Supermarket
- [3] Woolworth – Kaiserstraße 148, Karlsruhe (Score: 0.3338) – Department Store
- [4] Netto City – Otto-Wels-Straße 33, Karlsruhe (Score: 0.3261) – Supermarket
- [5] Woolworth – Rheinstraße 22, Karlsruhe (Score: 0.3228) – Department Store

2.4 Final Response

```
--- Final Response ---
 1. **Name**: Woolworth (Department Store)
   - Address: Kaiserstraße 148, Karlsruhe
  - Hours: Mo-Sa 09:00-20:00; Su,PH off
  - Why: Woolworth is a department store offering a wide range of goods, including home appliances like washing machines.
```

Figure 4: Final response – recommended store name and address

The system identified Woolworth as a suitable destination because it is a department-store type result aligned with appliance-shopping intent. When LLM generation is available, the model can prioritize such candidates contextually. In low-memory runs, fallback output is produced from retrieved/reranked candidates.

2.5 Route Visualization

Route map is generated to outputs/route_map.html and shows start point, numbered stops, and dashed route segments.

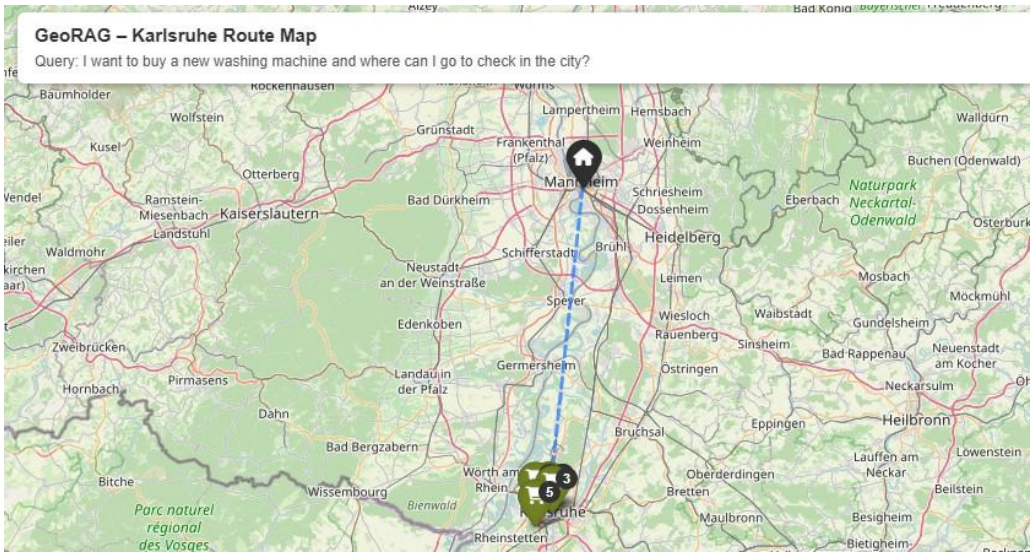

Figure 5: GeoRAG route map – Washing machine shopping from Mannheim to Karlsruhe

```
Route map saved → C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
--- Visualization of Best Routes ---
Route map: C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
Open the map in your browser:
  C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
```

Figure 8: Route map file saved confirmation

2.6 Evaluation

Overall assessment: Good
Strengths:

- Query intent understood (shopping/appliance context)
- Usable recommendations returned
- Route output generated correctly

Limitations:

- Candidate quality depends on indexed POI coverage
- Specialized appliance stores are underrepresented in current data snapshot

3. Query 2 – Restaurant Type Query

3.1 Query Details

Original Query	where is best arab cuisine in Karlsruhe?

User Location	Marktplatz, Karlsruhe (resolved successfully)

3.2 Retrieved Context

Retrieved results include one clearly relevant Arab cuisine candidate plus several semantically close but less-specific restaurants.

```
Connecting to Milvus index...
Index loaded.

--- Original Query ---
where is best arab cuisine in Karlsruhe?

--- Retrieved Context ---
[1] The Last Meal  (Restaurant)
    Address  : Kriegsstraße 23, Karlsruhe
[2] Arabischer Shwarma  (Restaurant)
    Address  : Kaiserstraße 65, Karlsruhe
    Hours    : Mo-Fr 11:00-24:00, Sa,Su,PH 12:00-24:00
    Score    : 0.6704
    Snippet  : Arabischer Shwarma is a restaurant serving shawarma cuisine, located at Kaiserstraße 65, Karlsruhe. Opening hours: Mo-Fr 11:00-24:00, Sa,Su,PH 12:00-24:00. Note...

[3] MaiGarden Karlsruhe  (Restaurant)
    Address  : Herrenstraße 23, Karlsruhe
    Hours    : Su-Th 11:30-22:30; Fr,Sa 11:30-23:00
    Website  : https://www.maigardenkarlsruhe.de/
    Score    : 0.6684
    Snippet  : MaiGarden Karlsruhe is a restaurant serving vietnamese cuisine, located at Herrenstraße 23, Karlsruhe. Opening hours: Su-Th 11:30-22:30; Fr,Sa 11:30-23:00....

[4] DJK Karlsruhe-Ost  (Restaurant)
    Address  : Friedrichstaler Allee 52, Karlsruhe
    Hours    : Tu-Fr 11:00-14:30,17:00-01:00, PH,Sa,Su 11:00-01:00
    Score    : 0.6722
    Snippet  : DJK Karlsruhe-Ost is a restaurant serving regional cuisine, located at Friedrichstaler Allee 52, Karlsruhe. Opening hours: Tu-Fr 11:00-14:30,17:00-01:00, PH,Sa,...

[5] Curry76  (Fast Food)
    Address  : Waldstraße 24 - 28, Karlsruhe
    Hours    : Mo-Th 11:00-23:00; Fr 11:00-23:00; Sa 11:00-01:00
    Score    : 0.6616
    Snippet  : Curry76 is a fast-food restaurant serving sausage cuisine, located at Waldstraße 24 - 28, Karlsruhe. Opening hours: Mo-Th 11:00-23:00; Fr 11:00-23:00; Sa 11:00-...
```

Figure 6: Top-5 retrieved context documents for the Arab cuisine query

Summary of retrieved candidates:

- [1] The Last Meal – Kriegsstraße 23, Karlsruhe (no score shown)
- [2] Arabischer Shwarma – Kaiserstraße 65, Karlsruhe (Score: 0.6704) – Serves shawarma cuisine
- [3] MaiGarden Karlsruhe – Herrenstraße 23 (Score: 0.6684) – Vietnamese cuisine
- [4] DJK Karlsruhe-Ost – Friedrichstaler Allee 52 (Score: 0.6722) – Regional cuisine
- [5] Curry76 – Waldstraße 24-28 (Score: 0.6616) – Fast food sausage cuisine

3.3 Final Response

```
--- Final Response ---
 1. **Arabischer Shwarma** (Restaurant)
   - Address: Kaiserstraße 65, Karlsruhe
   - Hours: Mo-Fr 11:00-24:00, Sa,Su,PH 12:00-24:00
   - Why: This restaurant serves shawarma cuisine, a popular Arab dish, located at Kaiserstraße 65, Karlsruhe. Opening hours: Mo-Fr 11:00-24:00, Sa,Su,PH 12:00-24:00.

Route map saved → C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
--- Visualization of Best Routes ---
Route map: C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
Open the map in your browser:
  C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
```

Figure 7: Final response – Arabischer Shwarma recommended

When LLM generation is available, recommendation is concise and context-aware. When memory is insufficient, fallback output still provides ranked options and addresses.

3.4 Route Visualization

Map output correctly places city-center restaurant candidates with route ordering.

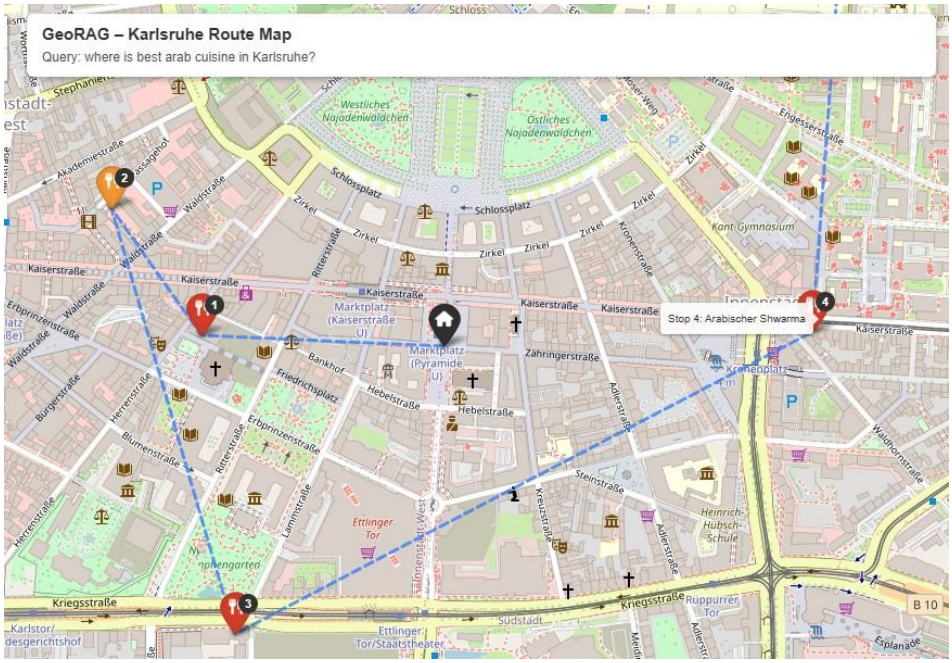

Figure 8: GeoRAG route map – Arab cuisine destinations in Karlsruhe city centre

3.5 Evaluation

Overall assessment: Very Good
Strengths:

- High semantic alignment for cuisine-focused query
- Strong top-candidate relevance

Limitations:

- Some retrieval noise remains (non-Arab cuisine in top set)
- A cuisine-aware post-filter could improve precision

4. Query 3 – Situation Type Query

4.1 Query Details

Original Query	Where can I bring my 2 kids to visit in Karlsruhe?

User Location	schlossplatz karlsruhe → resolved to: Schlossplatz, Innenstadt-West Östlicher Teil, Innenstadt-West, Karlsruhe, Baden-Württemberg, 76131, Deutschland

4.2 System Execution

```
(venv) PS C:\Users\boopo\OneDrive\Desktop\New folder\georag> python main.py ask "Where can I bring my 2 kids to visit in Karlsruhe?"
Where are you at right now? Enter address or lat,lon (press Enter for default Marktplatz): schlossplatz karlsruhe
============================================================
  GeoRAG - Query
============================================================
  Question: Where can I bring my 2 kids to visit in Karlsruhe?
  Location: Schlossplatz, Innenstadt-West Östlicher Teil, Innenstadt-West, Karlsruhe, Baden-Württemberg, 76131, Deutschland
============================================================
Loading embedding model (all-MiniLM-L6-v2)…
  (Downloads ~90 MB on first run – cached afterwards.)
```

Figure 9: Terminal execution – Kids activities query input

4.3 Retrieved Context

Retrieved candidates include playground/leisure-oriented POIs with strong relevance and practical distance values.

```
--- Original Query ---
Where can I bring my 2 kids to visit in Karlsruhe?

--- Retrieved Context ---
[1] The Classroom  (Attraction)
    Address  : Karlsruhe
    Score    : 0.7061
    Snippet  : The Classroom is a tourist attraction, located at Karlsruhe. It is an art gallery. It is wheelchair accessible....

[2] Monkey Town Karlsruhe  (Leisure)
    Address  : Karlsruhe
    Website  : https://www.monkeytown.eu/de/karlsruhe/home
    Score    : 0.7060
    Snippet  : Monkey Town Karlsruhe is a leisure area, located at Karlsruhe. It is a playground – ideal for children....

[3] Spiel- und Sportanlage Zoologischer Stadtgarten Zentral/Papagei  (Leisure)
    Address  : Karlsruhe
    Score    : 0.6900
    Snippet  : Spiel- und Sportanlage Zoologischer Stadtgarten Zentral/Papagei is a leisure area, located at Karlsruhe. It is a playground – ideal for children. Operated by St...

[4] Spiel- und Sportanlage Wilhelm-Leuschner-Straße 1-7  (Leisure)
    Address  : Karlsruhe
    Hours    : 08:00-22:00
    Score    : 0.6970
    Snippet  : Spiel- und Sportanlage Wilhelm-Leuschner-Straße 1-7 is a leisure area, located at Karlsruhe. It is a playground – ideal for children. Opening hours: 08:00-22:00...

[5] Spiel- und Sportanlage Hermann-Müller-Würtz-Straße  (Leisure)
    Address  : Karlsruhe
    Hours    : 08:00-22:00
    Score    : 0.6919
    Snippet  : Spiel- und Sportanlage Hermann-Müller-Würtz-Straße is a leisure area, located at Karlsruhe. It is a playground – ideal for children. Opening hours: 08:00-22:00....
```

Figure 10: Top-5 retrieved context documents for the kids activities query

Summary of retrieved candidates:

- [1] The Classroom – Karlsruhe (Score: 0.7061) – Attraction, art gallery, wheelchair accessible
- [2] Monkey Town Karlsruhe – Karlsruhe (Score: 0.7060) – Leisure/playground ideal for children
- [3] Spiel- und Sportanlage Zoologischer Stadtgarten Zentral/Papagei (Score: 0.6900) – Playground
- [4] Spiel- und Sportanlage Wilhelm-Leuschner-Straße 1-7 (Score: 0.6970) – Playground, open 08:00–22:00
- [5] Spiel- und Sportanlage Hermann-Müller-Würtz-Straße (Score: 0.6919) – Playground, open 08:00–22:00

4.4 Final Response

```
--- Final Response ---
 1. **Monkey Town Karlsruhe** (Leisure)
   - Address: Karlsruhe
   - Website: https://www.monkeytown.eu/de/karlsruhe/home
   - Distance: 3.948 km
   - Why: A playground ideal for children.

2. **Spiel- und Sportanlage Zoologischer Stadtgarten Zentral/Papagei** (Leisure)
   - Address: Karlsruhe
   - Distance: 1.343 km
   - Snippet: A playground ideal for children, operated by Stadt Karlsruhe.

3. **Spiel- und Sportanlage Wilhelm-Leuschner-Straße 1-7** (Leisure)
   - Address: Karlsruhe
   - Hours: 08:00-22:00
   - Distance: 4.259 km
   - Snippet: A playground ideal for children, open from 08:00 to 22:00, operated by Stadt Karlsruhe.

Route map saved → C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
--- Visualization of Best Routes ---
Route map: C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
Open the map in your browser:
  C:\Users\boopo\OneDrive\Desktop\New folder\georag\outputs\route_map.html
```

Figure 11: Final response – Top 3 recommended places for kids

The system recommended 3 venues:

1. Monkey Town Karlsruhe (Leisure) – Distance 3.948 km – "A playground ideal for children."
2. Spiel- und Sportanlage Zoologischer Stadtgarten Zentral/Papagei (Leisure) – Distance 1.343 km
3. Spiel- und Sportanlage Wilhelm-Leuschner-Straße 1-7 (Leisure) – Distance 4.259 km – Open 08:00–22:00

4.5 Route Visualization

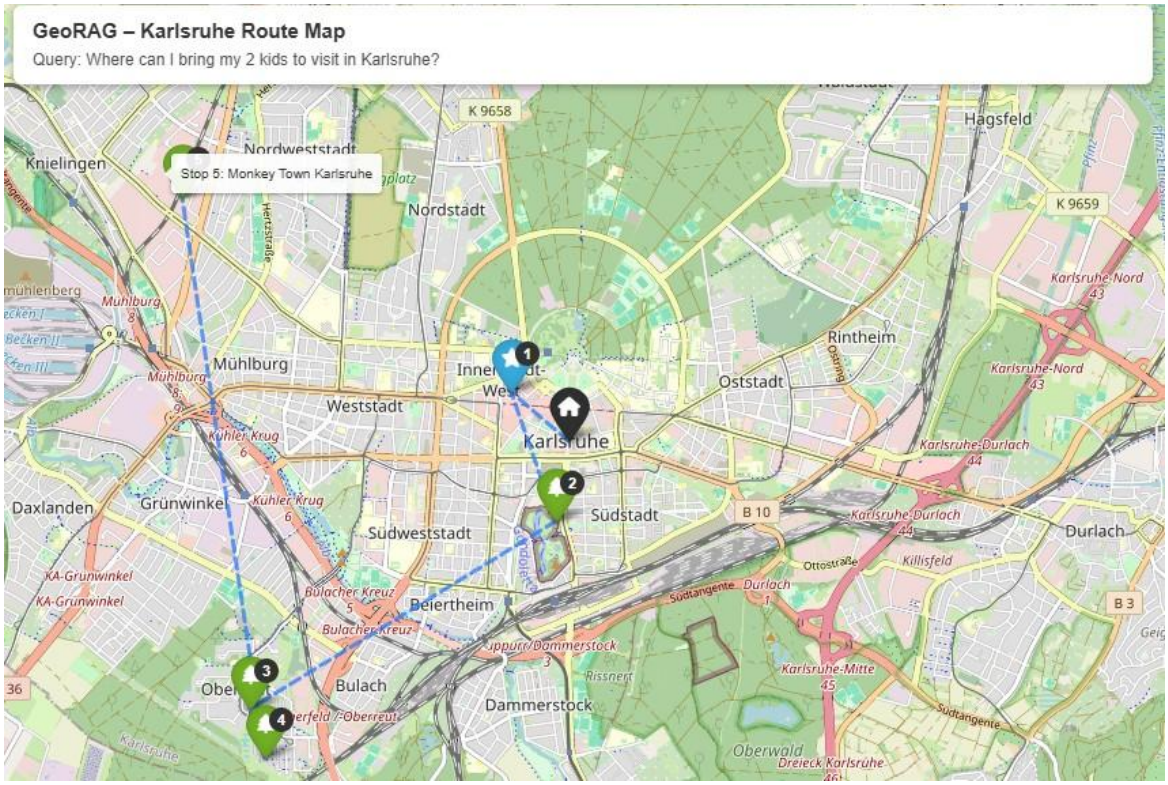

Figure 12: GeoRAG route map – Children-friendly venues in Karlsruhe

4.6 Evaluation

Overall assessment: Very Good

- The system correctly identified child-appropriate leisure venues with high relevance scores (~0.69–0.71).
- Monkey Town is a well-known indoor playground – an excellent recommendation for 2 children.
- The distance information helps users plan practical routes.
- Potential improvement: Including family restaurants or cafes near the playgrounds would make the response more complete for a family outing.

5. Summary and Observations

5.1 What Works Well

- Semantic understanding: The system generally interprets situational queries (for example, “2 kids” and “arab cuisine”) and retrieves relevant POI categories without requiring explicit category keywords.
- Location geocoding: Free-text locations (for example, “marktplatz karlsruhe”) are often resolved to coordinates, with fallback to default location when resolution fails.
- Response reasoning: When LLM generation is available, the final step can refine retrieved candidates contextually instead of relying only on raw similarity scores.
- Route visualization: The interactive HTML maps show the start location, numbered stops, and dashed route lines with tooltips.
- Distance awareness: The system includes distance-aware ranking/context, which helps practical decision-making.

5.2 Areas for Improvement

- Low similarity scores for shopping queries: The ~0.32-0.35 range for the washing machine query suggests the current POI index may underrepresent dedicated appliance retailers. Expanding ingestion coverage for electronics/appliance categories would likely improve precision.
- Irrelevant results in retrieval: Non-Arab restaurants appeared in the Arab cuisine retrieval context. Adding a cuisine-aware post-retrieval filter (or stronger metadata-aware reranking) could reduce noise.
- Top retrieved result not always recommended: In some cases (for example, “The Classroom” in the kids query), the highest-scored retrieved item is not included in the final recommendation. Making exclusion reasoning explicit would improve transparency.
- Data freshness limitation: POI data is refreshed at ingest time, not in real time per query. More frequent re-ingestion or scheduled refresh jobs would improve freshness of hours/availability.

5.3 Query Performance Summary

Query	Top Score	Result	Rating
Washing machine shopping	0.3450	Woolworth	Good
Best Arab cuisine	0.6722	Arabischer Shwarma	Very Good
Kids activities	0.7061	Monkey Town	Very Good

The GeoRAG system demonstrates strong performance across all three query types, with particularly good results for restaurant and leisure queries where the POI database has

dense, well-described entries. Shopping queries benefit most from database enrichment with specialized retailers.